The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP622
Subject Title	Emerging Memory Technologies
Credit Value	3
Level	6
Pre-requisite/ ~~Co-requisite/~~ ~~Exclusion~~	Undergraduate-level courses about electronic circuits, solid-state physics and semiconductor physics are required, e.g., AP20012, AP30011 and AP40006
Objectives	The objective of this subject is to introduce concepts of the memory sub-system from the device cell structures to the array and architecture design with emphasis on the industry trend and cutting-edge technologies. The concept of memory hierarchy is used as an outline through the whole course. The in-memory computing and in-sensor computing for artificial intelligence will be also covered.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Understand the working principles of conventional charge-based memory technology (b) Understand the working principles of emerging non-charge-based memory technology (c) Understand the fundamentals of in-memory computing and in-sensor computing (d) Understand the working principles of cross-point memory array for artificial neural network (e) Experimentally carry out the micro-fabrication and characterization of emerging memory using the facilities inside cleanroom
Subject Synopsis/ Indicative Syllabus	**Charge-based memory technology:** Static Random Access Memory, Dynamic Random Access Memory, Flash Memory, etc **Non-charge-based memory technology:** Resistive switching memory, phase-change memory, magnetic memory, etc **In-memory computing and in-sensor computing:** the principles of multiplication and accumulation **Cross-point memory array for artificial neural network**
Teaching/Learning Methodology	**Lecture**: Basic theory and knowledge will be systematically introduced in lectures. Class work and assignments related to the content of lectures will be used to enhance students learning.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
(1) Continuous assessment	50	√	√	√	√	√
(2) Project	50	√	√	√	√	√
Total	100					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Assignments will strengthen the students' basic knowledge and the analytical skills to solve the problems related to this subject. The course is a cutting edge technology-oriented course, thus the students are expected to actively read the related literature as part of the learning process. Hence, the proposed assessment methods are necessary to assess the intended learning outcomes.

Students should have gained the basic knowledge of a) charge-based memory technology and b) non-charge-based memory technology; c) have a thorough understanding of in-memory and in-sensor computing; and d) be able to understand the working principle of memory array for artificial neural network.

Student Study Effort Expected

Class contact:	
• Lecture/Seminar	21 h
Other student study effort:	
• Laboratory	18 h
• Self-study	81h
Total student study effort:	120 h

Reading List and References

1. Low Power and Reliable SRAM Memory Cell and Array Design, Authors: Koichiro Ishibashi, Kenichi Osada, Publisher: Springer, 2011.
2. DRAM Circuit Design: Fundamental and High-Speed Topics, 2nd edition, Authors: Brent Keeth, R. Jacob Baker, Brian Johnson, Feng Lin, Publisher: John Wiley & Sons, 2008
3. NAND FLASH Memory Technologies, Author: Seiichi Aritome, Publisher: Wiley-IEEE Press, 2016.
4. Resistive Random Access Memory (RRAM), Author: Shimeng Yu, Publisher: Morgan & Claypool, 2016.